

Introduction to C

Assignment 1

1. Accept radius of a circle and print the area and circumference of circle.
(Hint: $\text{area} = \pi r^2$, $\text{circumference} = 2\pi r$)



```
C pract_1.c
1 #include <stdio.h>
2
3 int main()
4 {
5     float r, area, circum;
6
7     printf("Enter radius: ");
8     scanf("%f", &r);
9
10    area = 3.14f * r * r;
11    circum = 2.0f * 3.14f * r;
12
13    printf("Area = %.2f\n", area);
14    printf("Circumference = %.2f\n", circum);
15
16    return 0;
17 }
18
19
20
```

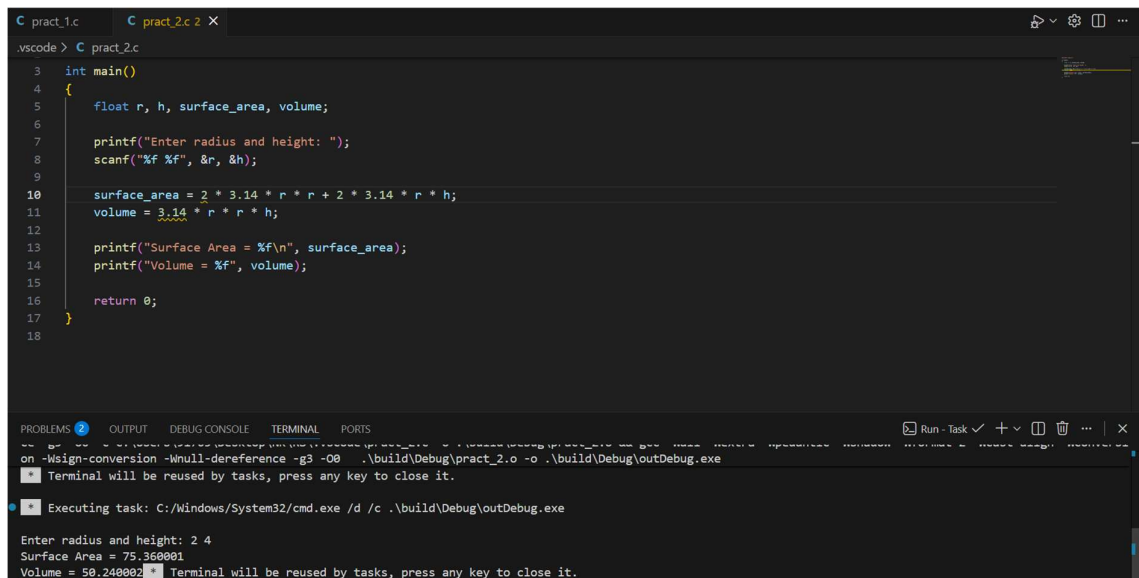
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter radius: 3
Area = 28.26
Circumference = 18.84

2. Accept dimensions of a cylinder and print the surface area and volume. (Hint: $\text{surface area} = 2\pi r^2 + 2\pi rh$, $\text{volume} = \pi r^2 h$)



```
C pract_1.c C pract_2.c X
vscode > C pract_2.c
3 int main()
4 {
5     float r, h, surface_area, volume;
6
7     printf("Enter radius and height: ");
8     scanf("%f %f", &r, &h);
9
10    surface_area = 2 * 3.14 * r * r + 2 * 3.14 * r * h;
11    volume = 3.14 * r * r * h;
12
13    printf("Surface Area = %f\n", surface_area);
14    printf("Volume = %f", volume);
15
16    return 0;
17 }
18
```

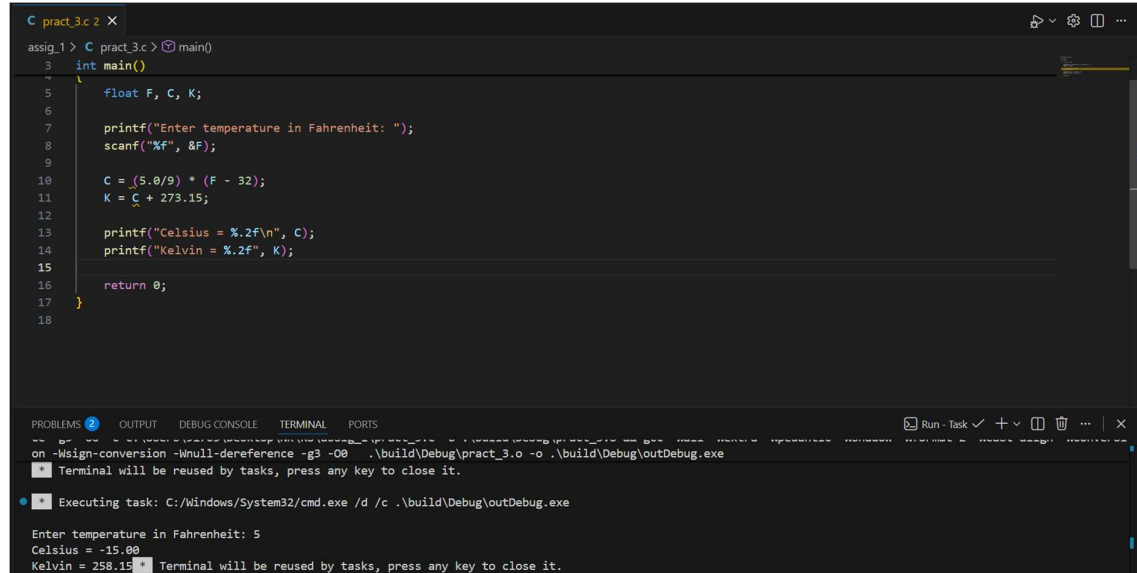
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter radius and height: 2 4
Surface Area = 75.360001
Volume = 50.240002

3. Accept temperatures in Fahrenheit(F) and print it in Celsius(C) and Kelvin (K).
(Hint: $C = 5/9(F - 32)$, $K = C + 273.15$)



```
C pract_3.c 2 X
assig_1 > C pract_3.c > main()
3 int main()
4 {
5     float F, C, K;
6
7     printf("Enter temperature in Fahrenheit: ");
8     scanf("%f", &F);
9
10    C = (5.0/9) * (F - 32);
11    K = C + 273.15;
12
13    printf("Celsius = %.2f\n", C);
14    printf("Kelvin = %.2f", K);
15
16    return 0;
17 }
18
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

on -Wsign-conversion -Wnull-dereference -g3 -O0 -.\build\Debug\pract_3.o -o .\build\Debug\outDebug.exe

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter temperature in Fahrenheit: 5
Celsius = -15.00
Kelvin = 258.15 Terminal will be reused by tasks, press any key to close it.

4. Accept initial velocity (u), acceleration (a) and time (t). Print the final velocity (v) and the distance (s) travelled. (Hint: $v = u + at$, $s = u + at^2$)



```
C pract_4.c X
assig_1 > C pract_4.c > main()
1 #include <stdio.h>
2
3 int main()
4 {
5     float u, a, t, v, s;
6
7     printf("Enter u, a and t: ");
8     scanf("%f %f %f", &u, &a, &t);
9
10    v = u + a * t;
11    s = u + a * t * t;
12
13    printf("Final Velocity = %.2f\n", v);
14    printf("Distance = %.2f", s);
15
16    return 0;
17 }
18
```

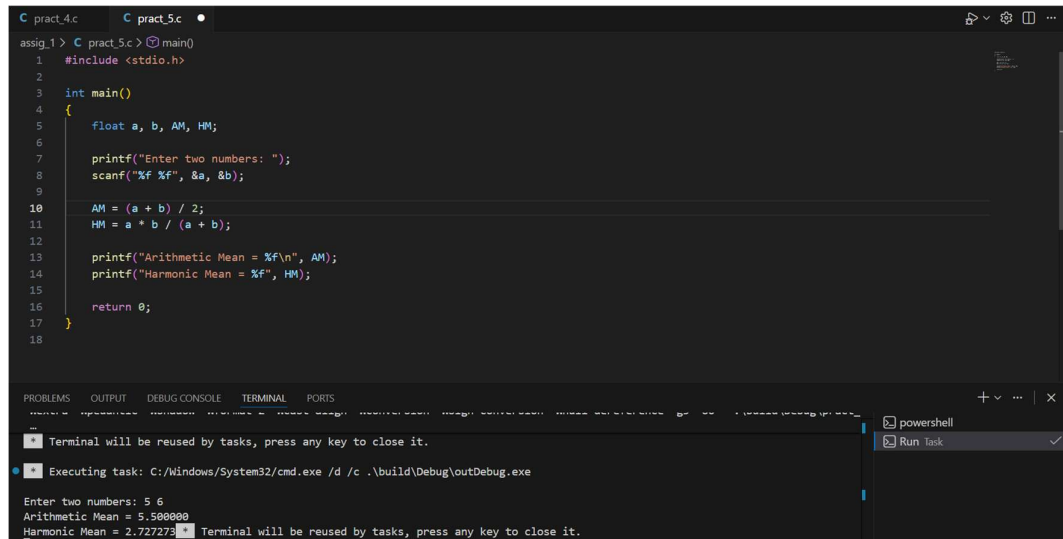
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter u, a and t: 5 4 3
Final Velocity = 17.00
Distance = 41.00 Terminal will be reused by tasks, press any key to close it.

5. Accept two numbers and print arithmetic and harmonic mean of the two numbers.
(Hint: $AM = (a+b)/2$, $HM = ab/(a+b)$)



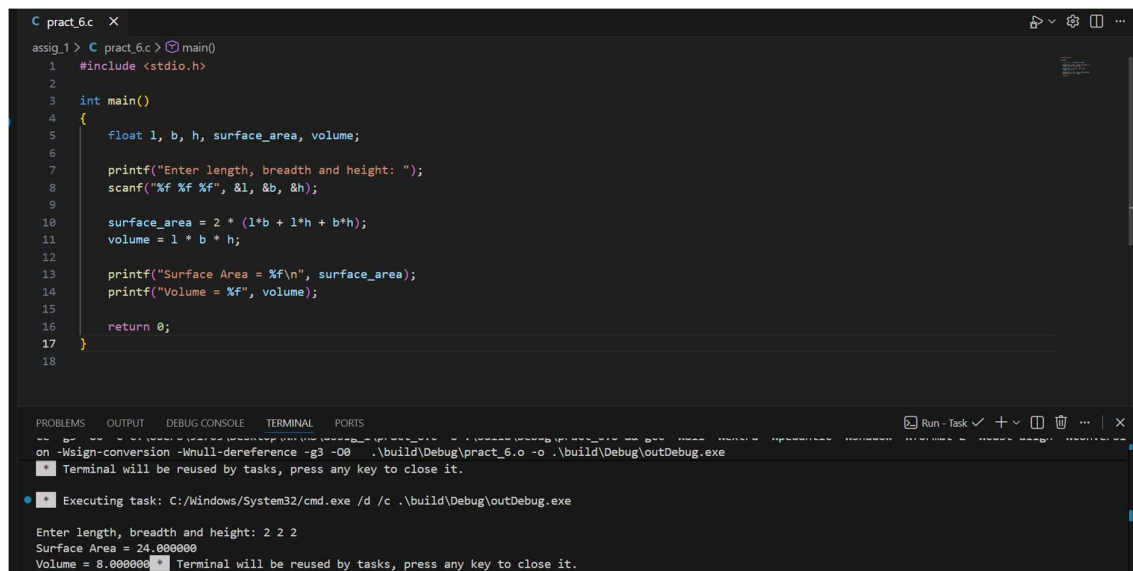
The screenshot shows a C program in a code editor. The program prompts the user to enter two numbers, calculates their Arithmetic Mean (AM) and Harmonic Mean (HM), and prints the results. The terminal output shows the user entering '5 6' and the program outputting 'Arithmetic Mean = 5.500000' and 'Harmonic Mean = 2.727273'.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float a, b, AM, HM;
6
7     printf("Enter two numbers: ");
8     scanf("%f %f", &a, &b);
9
10    AM = (a + b) / 2;
11    HM = a * b / (a + b);
12
13    printf("Arithmetic Mean = %f\n", AM);
14    printf("Harmonic Mean = %f", HM);
15
16    return 0;
17 }
```

Terminal Output:

```
Enter two numbers: 5 6
Arithmetic Mean = 5.500000
Harmonic Mean = 2.727273
```

6. Accept three dimensions length (l), breadth(b) and height(h) of a cuboid and print surface area and volume. (Hint: $surface\ area = 2(lb + lh + bh)$, $volume = lbh$)



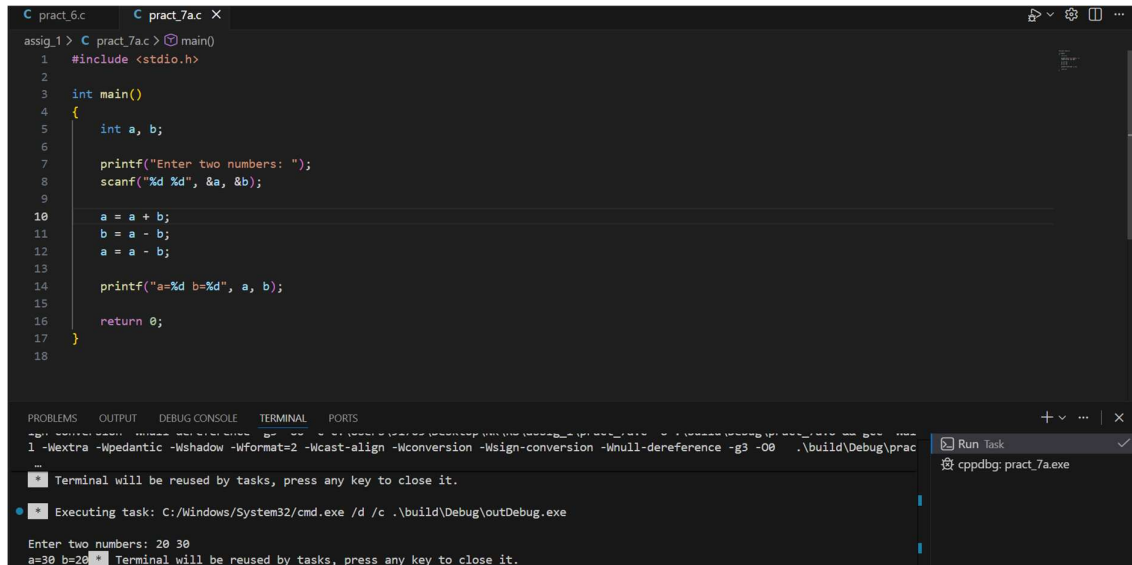
The screenshot shows a C program in a code editor. The program prompts the user to enter length, breadth, and height, calculates the surface area and volume, and prints the results. The terminal output shows the user entering '2 2 2' and the program outputting 'Surface Area = 24.000000' and 'Volume = 8.000000'.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float l, b, h, surface_area, volume;
6
7     printf("Enter length, breadth and height: ");
8     scanf("%f %f %f", &l, &b, &h);
9
10    surface_area = 2 * (l*b + l*h + b*h);
11    volume = l * b * h;
12
13    printf("Surface Area = %f\n", surface_area);
14    printf("Volume = %f", volume);
15
16    return 0;
17 }
```

Terminal Output:

```
Enter length, breadth and height: 2 2 2
Surface Area = 24.000000
Volume = 8.000000
```

7. Accept two integers from the user and interchange them. Display the interchanged numbers. Using temporary variable Without using temporary variable a. Using + and – operator b. Using / and * operator



```
C pract_7a.c X
assig_1 > C pract_7a.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      int a, b;
6
7      printf("Enter two numbers: ");
8      scanf("%d %d", &a, &b);
9
10     a = a + b;
11     b = a - b;
12     a = a - b;
13
14     printf("a=%d b=%d", a, b);
15
16     return 0;
17 }
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

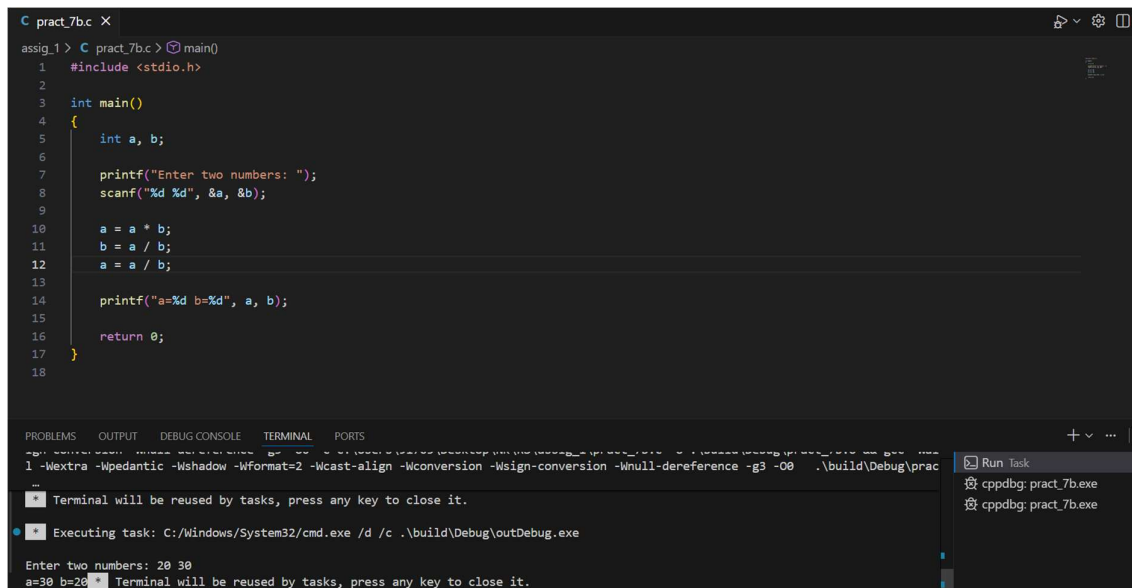
1 -Wextra -Wpedantic -Wshadow -Wformat=2 -Wcast-align -Wconversion -Wsign-conversion -Wnull-dereference -g3 -O0 .\build\Debug\prac

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter two numbers: 20 30
a=30 b=20

Run Task
c:\ppdbg\pract_7a.exe



```
C pract_7b.c X
assig_1 > C pract_7b.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      int a, b;
6
7      printf("Enter two numbers: ");
8      scanf("%d %d", &a, &b);
9
10     a = a * b;
11     b = a / b;
12     a = a / b;
13
14     printf("a=%d b=%d", a, b);
15
16     return 0;
17 }
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

1 -Wextra -Wpedantic -Wshadow -Wformat=2 -Wcast-align -Wconversion -Wsign-conversion -Wnull-dereference -g3 -O0 .\build\Debug\prac

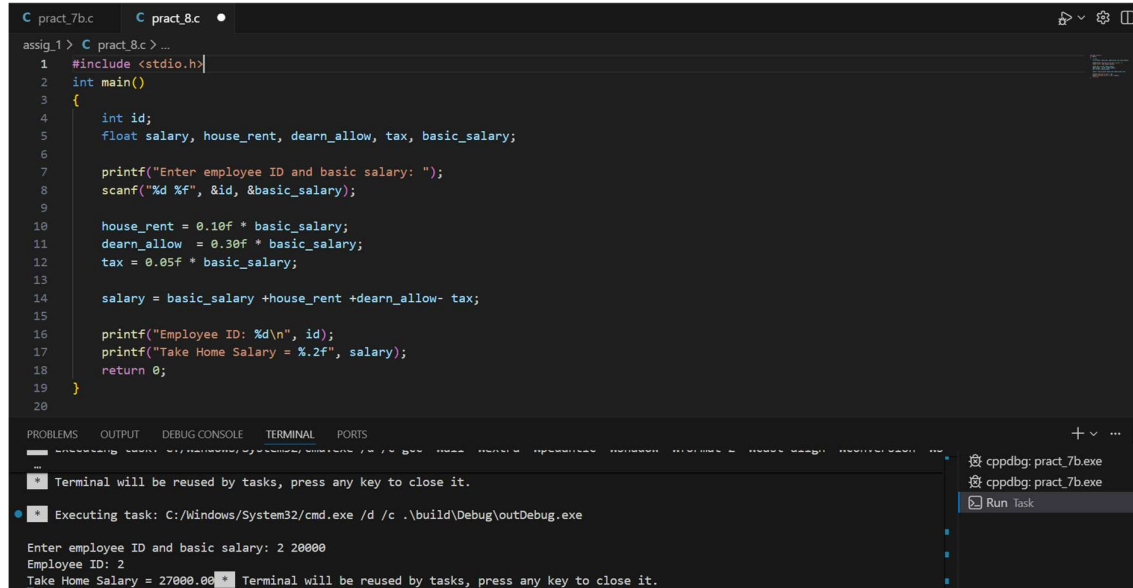
Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter two numbers: 20 30
a=30 b=20

Run Task
c:\ppdbg\pract_7b.exe
c:\ppdbg\pract_7b.exe

8. The basic salary of an employee is decided at the time of employment, which may be different for different employees. Apart from basic, employee gets 10% of basic as house rent, 30% of basic as dearness allowance. A professional tax of 5% of basic is deducted from salary. Accept the employee id and basic salary for an employee and output the take home salary of the employee.



```
1 #include <stdio.h>
2 int main()
3 {
4     int id;
5     float salary, house_rent, dearn_allow, tax, basic_salary;
6
7     printf("Enter employee ID and basic salary: ");
8     scanf("%d %f", &id, &basic_salary);
9
10    house_rent = 0.10f * basic_salary;
11    dearn_allow = 0.30f * basic_salary;
12    tax = 0.05f * basic_salary;
13
14    salary = basic_salary + house_rent + dearn_allow - tax;
15
16    printf("Employee ID: %d\n", id);
17    printf("Take Home Salary = %.2f", salary);
18    return 0;
19 }
20
```

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter employee ID and basic salary: 2 20000
Employee ID: 2
Take Home Salary = 27000.00

9. A cashier has currency notes of denomination 1, 5 and 10. Accept the amount to be withdrawn from the user and print the total number of currency notes of each denomination the cashier will have to give.



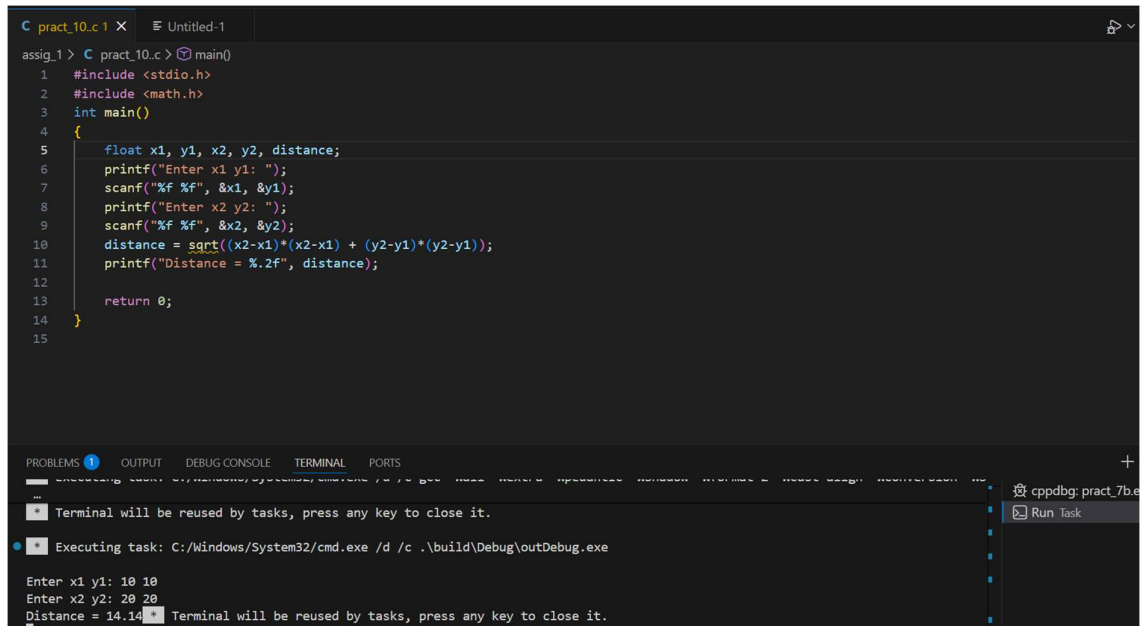
```
1 #include <stdio.h>
2 int main()
3 {
4     int amount;
5     int n1, n5, n10;
6     printf("enter amount: ");
7     scanf("%d", &amount);
8
9     n10 = amount / 10;
10    amount = amount % 10;
11    n5 = amount / 5;
12    amount = amount % 5;
13    n1 = amount;
14
15    printf("10 Rupee notes = %d\n", n10);
16    printf("5 Rupee notes = %d\n", n5);
17    printf("1 Rupee notes = %d", n1);
18    return 0;
19 }
20
```

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

enter amount: 1234
10 Rupee notes = 123
5 Rupee notes = 0
1 Rupee notes = 4

10. Accept the x and y coordinates of two points and compute the distance between the two points.

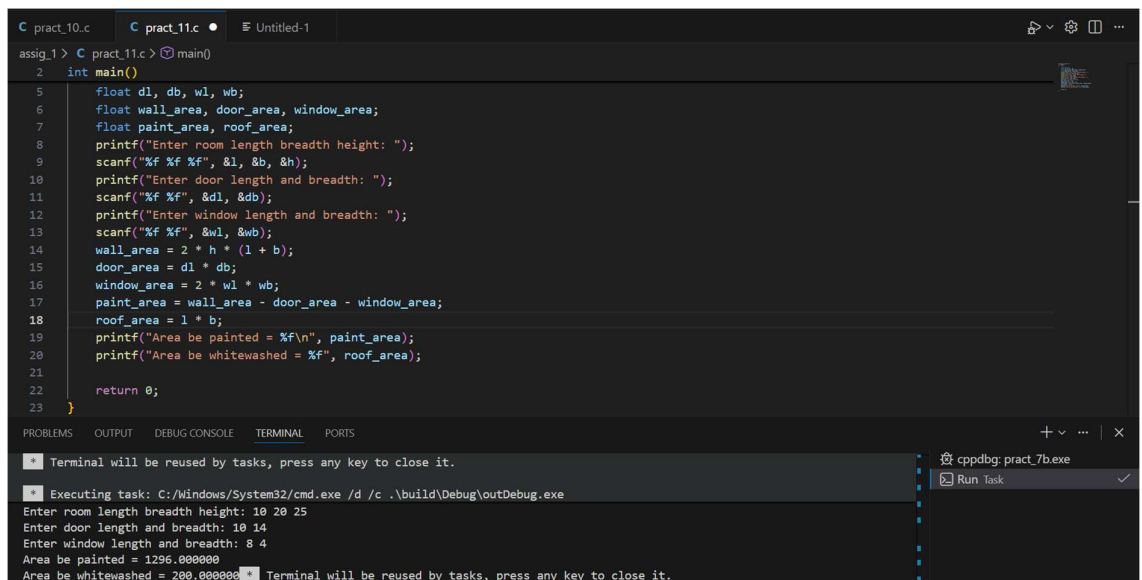


```
C pract_10.c X  Untitled-1
assig_1 > C pract_10.c > main()
1  #include <stdio.h>
2  #include <math.h>
3  int main()
4  {
5      float x1, y1, x2, y2, distance;
6      printf("Enter x1 y1: ");
7      scanf("%f %f", &x1, &y1);
8      printf("Enter x2 y2: ");
9      scanf("%f %f", &x2, &y2);
10     distance = sqrt((x2-x1)*(x2-x1) + (y2-y1)*(y2-y1));
11     printf("Distance = %.2f", distance);
12
13     return 0;
14 }
15

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Terminal will be reused by tasks, press any key to close it.
Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter x1 y1: 10 10
Enter x2 y2: 20 20
Distance = 14.14
```

11. Consider a room having one door and two windows both of the same size. Accept dimensions of the room, door and window. Print the area to be painted (interior walls) and area to be whitewashed (roof).

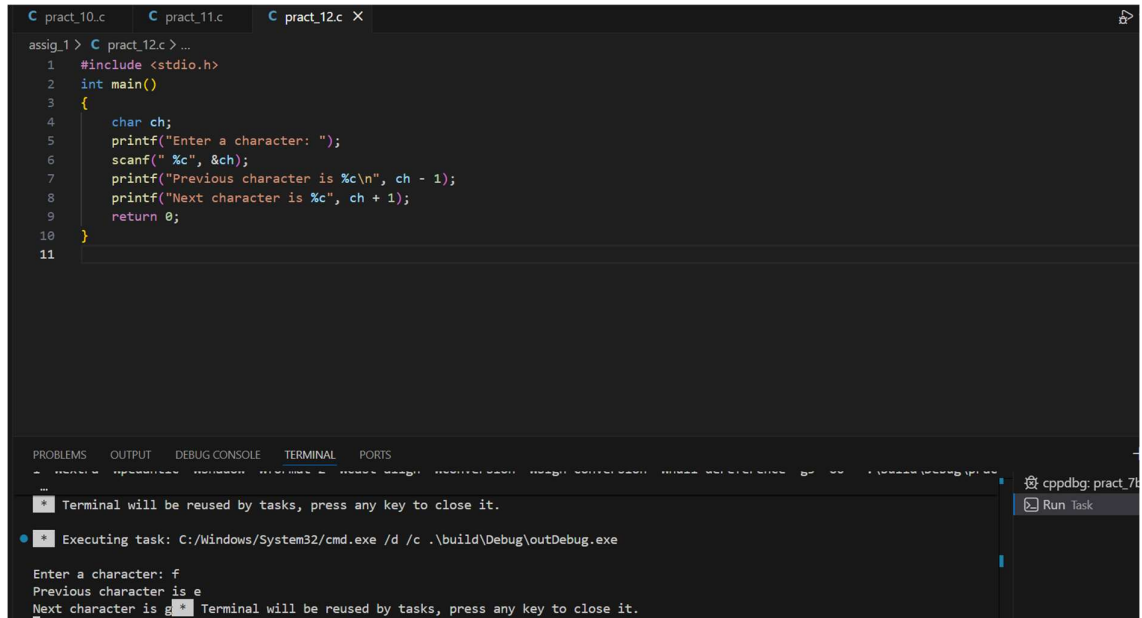


```
C pract_10.c C pract_11.c  Untitled-1
assig_1 > C pract_11.c > main()
2  int main()
3  {
4      float dl, db, wl, wb;
5      float wall_area, door_area, window_area;
6      float paint_area, roof_area;
7      printf("Enter room length breadth height: ");
8      scanf("%f %f %f", &l, &b, &h);
9      printf("Enter door length and breadth: ");
10     scanf("%f %f", &dl, &db);
11     printf("Enter window length and breadth: ");
12     scanf("%f %f", &wl, &wb);
13     wall_area = 2 * h * (l + b);
14     door_area = dl * db;
15     window_area = 2 * wl * wb;
16     paint_area = wall_area - door_area - window_area;
17     roof_area = l * b;
18     printf("Area be painted = %f\n", paint_area);
19     printf("Area be whitewashed = %f", roof_area);
20
21     return 0;
22 }
23

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Terminal will be reused by tasks, press any key to close it.
Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter room length breadth height: 10 20 25
Enter door length and breadth: 10 14
Enter window length and breadth: 8 4
Area be painted = 1296.000000
Area be whitewashed = 200.000000
```

12. Accept a character from the keyboard and display its previous and next character in order. Ex. If the character entered is 'd', display "The previous character is c", "The next character is e".



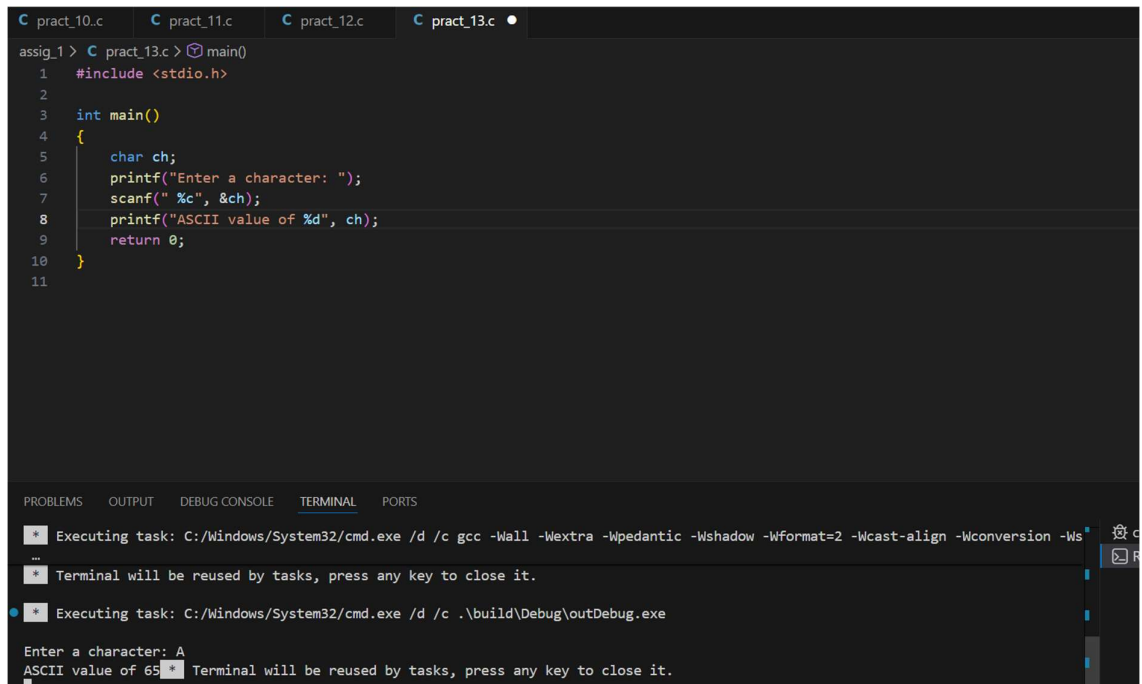
```
assig_1 > C pract_12.c > ...
1  #include <stdio.h>
2  int main()
3  {
4      char ch;
5      printf("Enter a character: ");
6      scanf(" %c", &ch);
7      printf("Previous character is %c\n", ch - 1);
8      printf("Next character is %c", ch + 1);
9      return 0;
10 }
11
```

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter a character: f
Previous character is e
Next character is g

13. Accept a character from the user and display its ASCII value



```
assig_1 > C pract_13.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      char ch;
6      printf("Enter a character: ");
7      scanf(" %c", &ch);
8      printf("ASCII value of %d", ch);
9      return 0;
10 }
11
```

Executing task: C:/Windows/System32/cmd.exe /d /c gcc -Wall -Wextra -Wpedantic -Wshadow -Wformat=2 -Wcast-align -Wconversion -W...

Terminal will be reused by tasks, press any key to close it.

Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\outDebug.exe

Enter a character: A
ASCII value of 65